

Optimising using corner points principle



1

- a
- i 14
 - ii 39
 - iii 54
 - iv 44
- b The maximum value of the objective function is 54 at the point (6, 10).
- c The minimum value of the objective function is 14 at the point (2, 2).

2

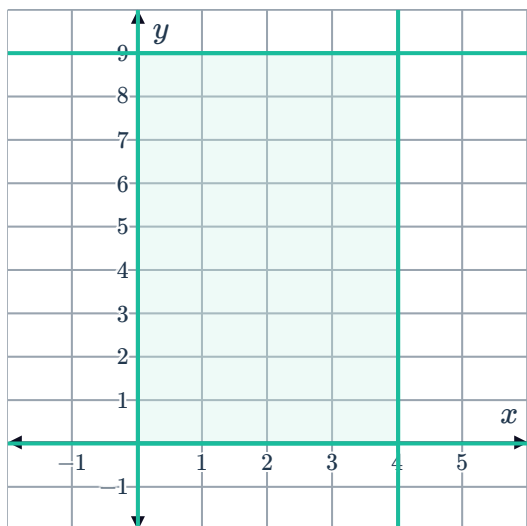
- a
- i The maximum value of the objective function is 48 at the point (6, 9).
 - ii The minimum value of the objective function is 14 at the point (2, 2).
- b
- i The maximum value of the objective function is 87 at the point (9, 10).
 - ii The minimum value of the objective function is 15 at the point (1, 2).
- c
- i The maximum value of the objective function is 51 at the point (8, 9).
 - ii The minimum value of the objective function is 12 at the point (2, 2).
- d
- i The maximum value of the objective function is 90 at the point (3, 9).
 - ii The minimum value of the objective function is 0 at the point (3, 0).
- e
- i The maximum value of the objective function is 35 at the point (9, 5)
 - ii The minimum value of the objective function is -10 at the point (2, 10).

3 The maximum value of the objective function T is 279 at the point (0, 9).

4

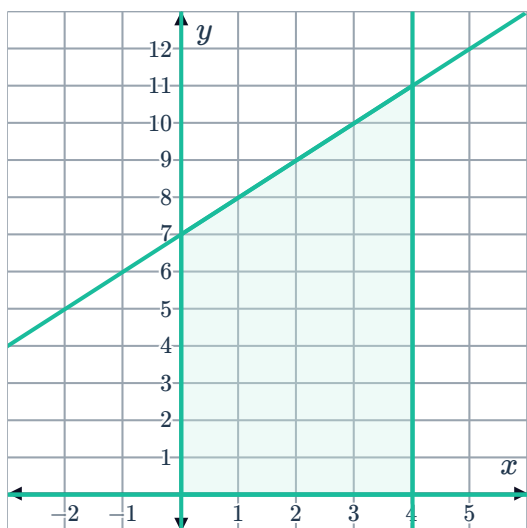
a

i

ii $(0,0), (4,0), (4,9), (0,9)$

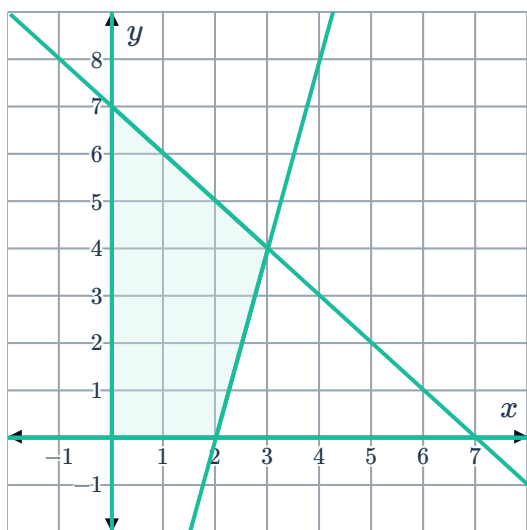
b

i

ii $(0,0), (4,0), (4,11), (0,7)$

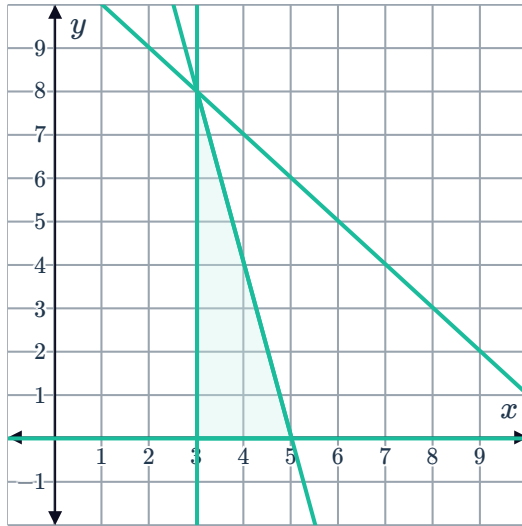
c

i

ii $(0,0), (2,0), (3,4), (0,7)$

d

i



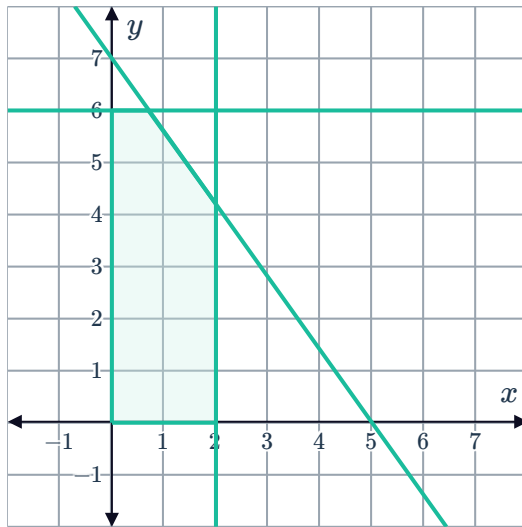
ii

$(3,0), (5,0), (3,8)$

5

a

i



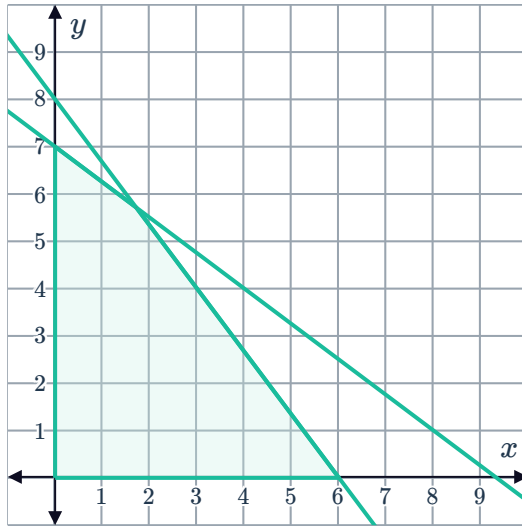
ii

The maximum is 116 at the point $(0,6)$.

iii The minimum is 0 at the point $(2,0)$

b

i



ii

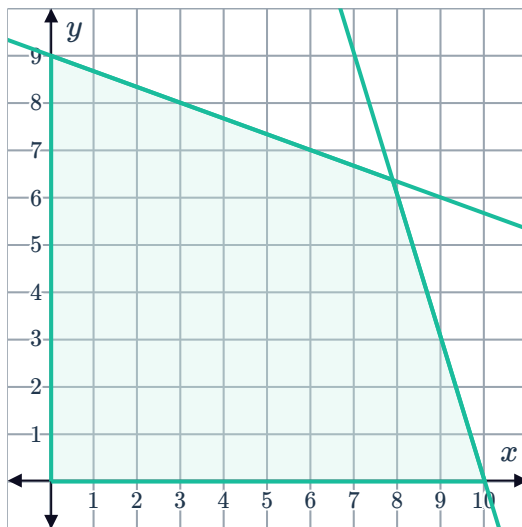
The maximum is 322 at the point $(0, 7)$.

iii The minimum is 0 at the point $(0, 0)$.

6

a

i



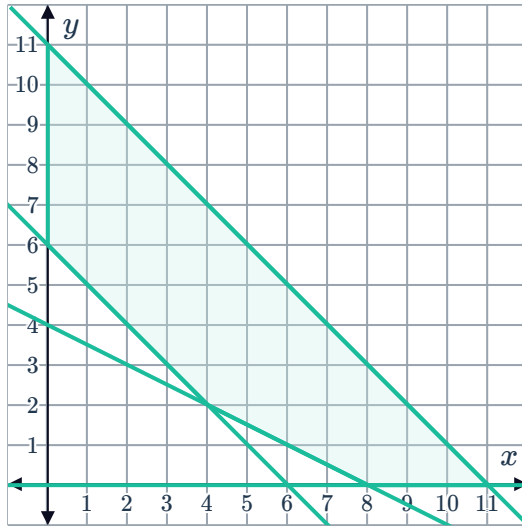
ii

The maximum value of P is 60 at $(10, 0)$.

iii No solution provided.

b

i



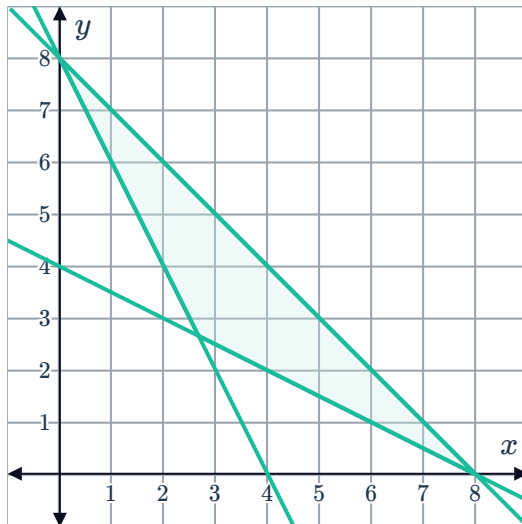
ii

The maximum value of z is 242 at $(11, 0)$.

iii No solution provided

c

i



ii

The maximum value of z is 112 at $(0, 8)$.

iii No solution provided



Applications

7

a $x \geq 0, y \geq 0, 2x + 3y \leq 36, 2x + 2y \leq 32$

b



c $(0, 0), (0, 12), (12, 4), (16, 0)$

d $P = 240x + 280y$

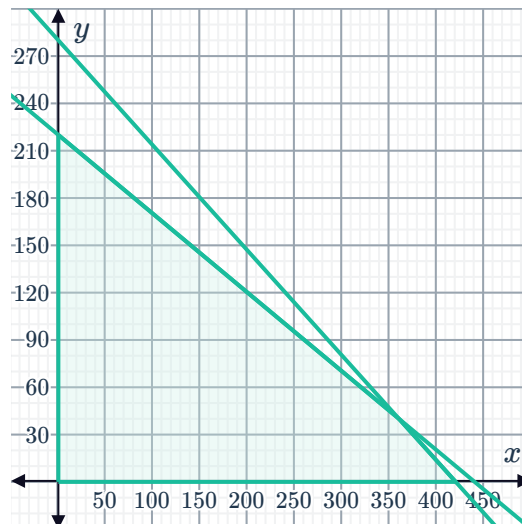
e $(12, 4)$

f \$4000

8

a $x \geq 0, y \geq 0, 8x + 16y \leq 3520$ and $8x + 12y \leq 3360$

b



c $(0, 0), (0, 220), (360, 40), (420, 0)$

d $P = 15x + 26y$

e $(360, 40)$

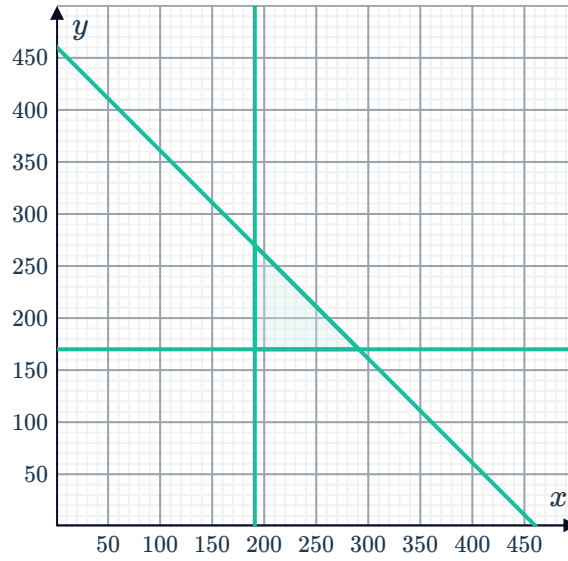
f \$6640

9



a $x \geq 190, y \geq 170$ and $x + y \leq 460$

b



c $P = 1.70x + 1.00y$

d

Vertex	Profit (\$)
(190, 170)	493
(190, 270)	593
(290, 170)	663

e \$663

f 290 beef and 170 chicken burgers.

g 190 beef and 270 chicken burgers.